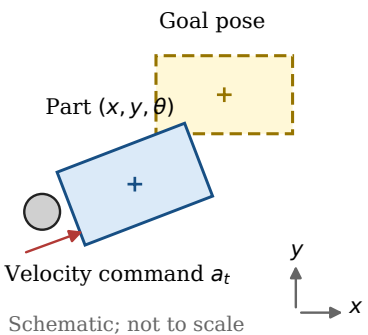
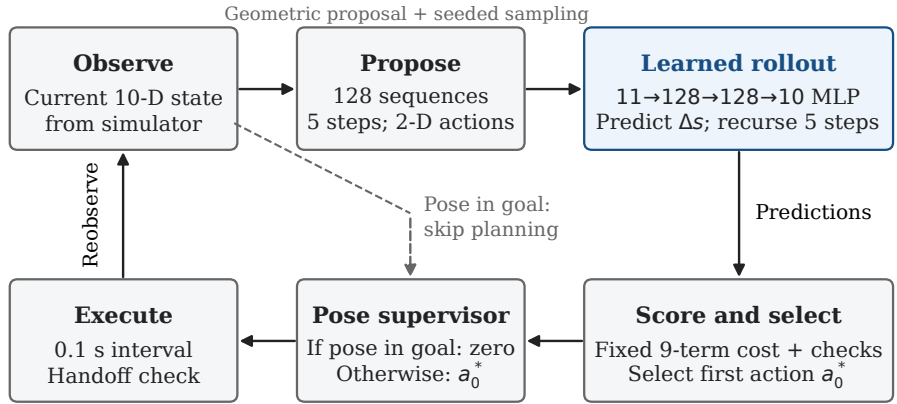
(a) Planar alignment task



(b) S1 / SK learned-dynamics planning loop



MLP: multilayer perceptron. S1: one-step loss; SK: one-step + 0.1 × 5-step free-rollout loss.

R: feature-linear model (α = 0) replaces the MLP. G: direct geometric feedback; no candidate search.

128 candidates; 5 × 0.1 s prediction steps; execute only the first command. Supervisor and contact physics are not learned.